

Involuntary Exhalation Control by Facial Vibration

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ABSTRACT

Breathing affects physical and mental health as well as skills in playing sports and musical instruments. Previous studies proposed various methods using sensory stimuli to assist users with controlling their breathing voluntarily by paying attention to it. However, focusing on breathing is difficult when they play sports or instruments because there are many factors to focus on. Therefore, we propose a wearable system that can control the user's exhalation involuntarily by facial vibration; pushing air from the cheeks independent of the user's voluntary breathing. Our system can control the exhaled air velocity and duration by changing the frequency, amplitude, and duration of the facial vibration. We consider our system will help novices acquire advanced skills in playing wind instruments like circular breathing.

CCS CONCEPTS

• Human-centered computing → Haptic devices.

KEYWORDS

breathing, involuntary exhalation control, facial vibration

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1 INTRODUCTION

Breathing affects both physical and mental health such as the treatment of hypertensive patients [8] and asthma patients [11], and reducing stress [12]. In addition, controlling breathing is an important skill in playing sports [1] and wind instruments [2]. Therefore, assisting systems to control breathing properly have the potential to maintain physical and mental health as well as help users achieve high performance in playing sports and instruments.

Previous studies used sensory stimuli such as visual [7, 10, 13], auditory [3, 10, 13], and tactile [4, 6] stimuli to assist users with changing their breathing; users change the rhythms and depth of breathing voluntarily depending on presented sensory stimuli. When using these systems, users need to pay attention to their

breathing. However, focusing on breathing is difficult when they play sports or instruments because there are many factors to focus on. For example, when playing wind instruments, players need to be aware of fingering, scores, and music as well as breathing. Therefore, we should consider an involuntary breathing control system that does not require the user's voluntary control of breathing, i.e., inducing inhalation or exhalation independent of the user's voluntary breathing.

In this study, we propose a wearable system that can control involuntary exhalation by facial vibration. We use two vibration speakers attached to the cheeks; pushing air from the cheeks by facial vibration on the basis of the principle of an air cannon with vibration speakers [9]. Our prototype system (Figure 1) can control the exhaled air velocity and duration by changing the frequency, amplitude, and duration of the facial vibration; there is no need for users to change their breathing voluntarily.

2 IMPLEMENTATION

This section describes the hardware, software, and algorithm of our system and a conducted pilot study.

2.1 Hardware

Our system consists of two vibration speakers (Vibro-Transducer Vp210, Acoupe Laboratory) attached to headphones, an amplifier (LP-268, Lepy), and a laptop. Users wear the headphones on their cheeks as shown in Figure 1 (b) to position the vibration speakers. The laptop generates pulse waves, the low-frequency waves are amplified by the amplifier, and the vibration is provided through the vibration speakers.

2.2 Software

The vibration is generated from a chirp signal of pulse waves that decreases the frequency linearly within a few seconds, e.g., 80 Hz to 1 Hz within two seconds. That is, the intensity of vibration is at the highest at the start and lowest at the end. We used this chirp signal because the necessary intensity to puff out air from the cheeks depends on the air volume within the cheeks; we would not provide stronger vibration than necessary that causes discomfort for the user.

2.3 Algorithm

Our system can control the exhaled air velocity and duration by changing the frequency, amplitude, and duration of the facial vibration. Due to the principle of our system that pushes air from the cheeks, the exhaled air volume also depends on the air volume in the mouth, i.e., whether the cheeks are inflated or not.

The relationship between these parameters is as follows:

*Both authors contributed equally to this research.

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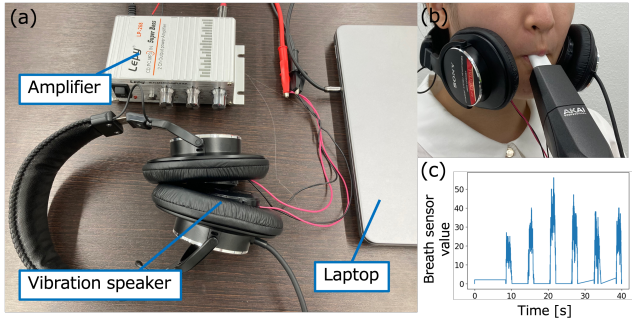


Figure 1: Overview of our prototype. (a) It consists of two vibration speakers attached to headphones, an amplifier, and a laptop. (b) How to wear our device. User can play a wind synthesizer with it. (c) Example of measured breath sensor values with a wind synthesizer. Each pulse represents involuntary exhalation by facial vibration for 2 s.

- (1) The higher the vibration frequency or amplitude, the higher the exhaled air velocity.
- (2) The longer the vibration duration, the longer the exhalation duration.
- (3) The higher the air volume in the mouth, the higher the exhaled air volume.

2.4 Pilot study

We measured the exhaled air velocity, duration, and volume puffed out by facial vibration in the following procedure. A participant, one of the authors, wore the vibration speakers on their cheeks with a mask and held a wind synthesizer (EWI Solo, Akai) in their mouth; the wind synthesizer is equipped with a breath sensor that can measure exhaled air velocity as A/D conversion values of 0–127.

2.4.1 Procedure.

- (1) The participant held the wind synthesizer in their mouth with holding their breath; the breath sensor value was 0.
- (2) The vibration was presented under various conditions by changing the frequency, amplitude, and duration of the vibration and the air volume in the mouth; the breath sensor value changed depending on the exhaled air by the facial vibration.
- (3) We repeated the procedure five times under each condition and calculated the median value for each parameter.

Figure 1 (c) shows an example of breath sensor values. We calculated the average value as the exhaled air velocity, the integration value as the exhaled air volume for each exhalation, and the duration from the rising to falling as the exhalation duration.

2.4.2 Conditions. We set the default vibration parameters as follows: the frequency of the chirp was 80–1 Hz, the amplitude was 1 (max), the duration was 2 s, and the air volume in the mouth was maximum (100 %) by the participant inflating their cheeks. Then we changed one of the parameters as follows while maintaining the other parameters.

- Frequency at the start: every 20 Hz from 20 Hz to 100 Hz.

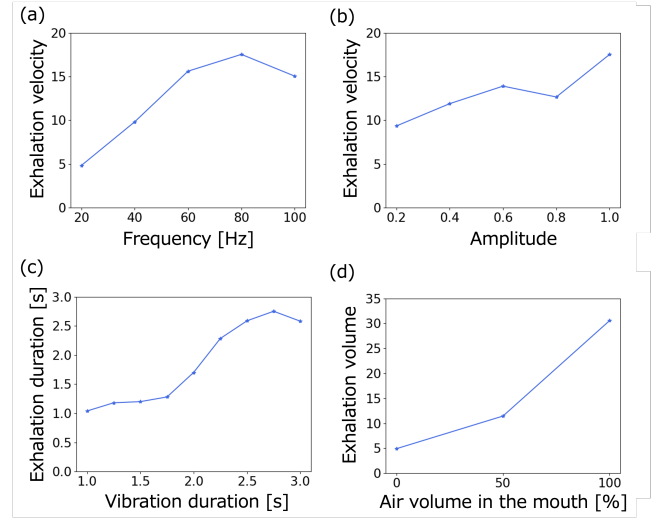


Figure 2: Results in the pilot study under each condition: (a) frequency-exhalation velocity, (b) amplitude-exhalation velocity, (c) vibration duration-exhalation duration, (d) air volume in the mouth-exhalation volume. Calculated median values among five trials under each condition are plotted.

- Amplitude: every 0.2 from 0.2 to 1.0.
- Duration: every 0.25 s from 1.0 s to 3.0 s.
- Air volume in the mouth: 0 %, 50 %, or 100 %

2.4.3 Results. Figure 2 shows the results under each condition. These results suggested that our prototype could control the exhaled air velocity and duration involuntarily by changing the vibration parameters on the basis of our algorithm; increasing the frequency or amplitude induced higher exhaled air velocity, increasing the vibration duration induced longer exhalation duration, and increasing the air volume in the mouth induced more exhaled air volume.

3 DISCUSSION

We explored an involuntary breathing control system that does not require the user's voluntary control of breathing; induces exhalation independent of the user's voluntary breathing. We consider one of the use cases with our system; it would help users acquire advanced skills in playing wind instruments like circular breathing [5]. Circular breathing requires players to exhale through the mouth and inhale through the nose at the same time, and the technique is not easy for novices. When novices practiced with our system, they would do it easily by focusing on the inhalation through the nose because they can exhale through the mouth involuntarily by facial vibration. We believe involuntary breathing control systems will help novices learn advanced skills in playing several sports and instruments.

The limitations are as follows. First, the controllable breathing is limited to exhalation. Second, the induced exhalation is not related to respiration, i.e., the air is not from the lungs. For the purpose of improving physical and mental health by involuntary breathing,

we have to solve these limitations. Thus, we will consider other methods to induce involuntary inhalation and exhalation from the lungs in future work.

REFERENCES

- [1] Felipe Contreras-Briceño, Maximiliano Espinosa-Ramírez, Eduardo Moya-Gallardo, Rodrigo Fuentes-Kloss, Luigi Gabrielli, Oscar F. Araneda, and Ginés Viscor. 2021. Intercostal Muscles Oxygenation and Breathing Pattern during Exercise in Competitive Marathon Runners. *International Journal of Environmental Research and Public Health* 18, 16 (Aug. 2021), 8287. <https://doi.org/10.3390/ijerph18168287>
- [2] Isabelle Cossette, Pawel Sliwinski, and Peter T. Macklem. 2000. Respiratory parameters during professional flute playing. *Respiration Physiology* 121, 1 (June 2000), 33–44. [https://doi.org/10.1016/S0034-5687\(00\)00111-0](https://doi.org/10.1016/S0034-5687(00)00111-0)
- [3] Jason Harris, Sarah Vance, Odair Fernandes, Avinash Parnandi, and Ricardo Gutierrez-Osuna. 2014. Sonic respiration: controlling respiration rate through auditory biofeedback. In *CHI '14 Extended Abstracts on Human Factors in Computing Systems*. ACM, Toronto Ontario Canada, 2383–2388. <https://doi.org/10.1145/2559206.2581233>
- [4] Annkatrin Jung, Miquel Alfaras, Pavel Karpashevich, William Primett, and Kristina Höök. 2021. Exploring Awareness of Breathing through Deep Touch Pressure. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. ACM, Yokohama Japan, 1–15. <https://doi.org/10.1145/3411764.3445533>
- [5] Angus McPherson and Carolyn Philpott. 2016. Circular Breathing: Expanding Musical Possibilities for Flute Players and Composers. *Journal of Music Research Online* 7 (2016), 1–11.
- [6] Pardis Miri, Emily Jusuf, Andero Uusberg, Horia Margarit, Robert Flory, Katherine Isbister, Keith Marzullo, and James J. Gross. 2020. Evaluating a Personalizable, Inconspicuous Vibrotactile (PIV) Breathing Pacer for In-the-Moment Affect Regulation. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*. ACM, Honolulu HI USA, 1–12. <https://doi.org/10.1145/3313831.3376757>
- [7] Neema Moraveji, Athman Adiseshan, and Takehiro Hagiwara. 2012. BreathTray: augmenting respiration self-regulation without cognitive deficit. In *CHI '12 Extended Abstracts on Human Factors in Computing Systems*. ACM, Austin Texas USA, 2405–2410. <https://doi.org/10.1145/2212776.2223810>
- [8] Monika Mourya, Aarti Sood Mahajan, Narinder Pal Singh, and Ajay K. Jain. 2009. Effect of Slow- and Fast-Breathing Exercises on Autonomic Functions in Patients with Essential Hypertension. *The Journal of Alternative and Complementary Medicine* 15, 7 (July 2009), 711–717. <https://doi.org/10.1089/acm.2008.0609>
- [9] Rajinder Sodhi, Ivan Poupyrev, Matthew Glisson, and Ali Israr. 2013. AIREAL: interactive tactile experiences in free air. *ACM Transactions on Graphics* 32, 4 (July 2013), 1–10. <https://doi.org/10.1145/2461912.2462007>
- [10] Tobias Sonne and Mads Møller Jensen. 2016. ChillFish: A Respiration Game for Children with ADHD. In *Proceedings of the TEI '16: Tenth International Conference on Tangible, Embedded, and Embodied Interaction*. ACM, Eindhoven Netherlands, 271–278. <https://doi.org/10.1145/2839462.2839480>
- [11] M Thomas. 2003. Breathing retraining for dysfunctional breathing in asthma: a randomised controlled trial. *Thorax* 58, 2 (Feb. 2003), 110–115. <https://doi.org/10.1136/thorax.58.2.110>
- [12] Liza Varvogli and Christina Darviri. 2011. Stress Management Techniques: evidence-based procedures that reduce stress and promote health. *HEALTH SCIENCE JOURNAL* 5, 2 (2011), 16.
- [13] Bin Yu, Jun Hu, Mathias Funk, Rong-Hao Liang, Mengru Xue, and Loe Feijs. 2018. RESonance: Lightweight, Room-Scale Audio-Visual Biofeedback for Immersive Relaxation Training. *IEEE Access* 6 (2018), 38336–38347. <https://doi.org/10.1109/ACCESS.2018.2853406>